

Introduction to Quantum Computing

(Session 1 – Day 1, 3 July 2026)

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Learning Objectives

- Understand why quantum mechanics offers a computational advantage.
- Grasp the basic postulates: state space, evolution, measurement.
- Visualise a qubit on the Bloch sphere.
- Work through a simple toy example (Hadamard + measurement) analytically and numerically.
- Recognise the difference between classical bits and quantum bits (qubits).
- Understand quantum complexity classes (**BQP**, **QMA**) and their relations to classical classes (**P**, **BPP**, **NP**, **PSPACE**).
- Comprehend the three major paradigms of quantum computing (Circuit Model, Measurement-Based, and Adiabatic) and their equivalence in computational power.
- Compare different physical platforms for implementing qubits (superconducting, trapped ions, neutral atoms, photonics, and spin qubits) based on their physical mechanisms, advantages, and key challenges.

1 Why Quantum Computing?

- Certain problems (such as integer factoring and simulation of quantum systems) admit polynomial-time quantum algorithms, whereas unstructured search admits a quadratic quantum speedup.
- Rather than a single source of speedup, quantum computation exploits superposition, interference, and, in many multi-qubit settings, entanglement to perform computations.

2 Postulates of Quantum Mechanics (Qubit-Centric View)

1. **State Space:** The state of an isolated quantum two-level system (qubit) is a unit vector in a complex Hilbert space $\mathbb{C}^2$:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1,$$

where $|0\rangle, |1\rangle$ form an orthonormal basis (the *computational basis*).

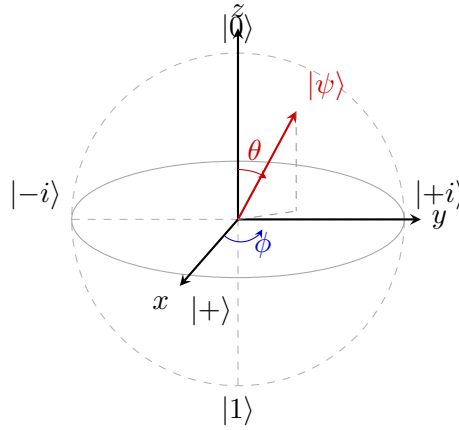
2. **Evolution:** Closed-system dynamics are unitary: $|\psi(t)\rangle = U(t) |\psi(0)\rangle$ with $U^\dagger U = I$. Elementary gates are 2×2 unitary matrices (Pauli- X, Y, Z , Hadamard H , phase S, T , etc.).
3. **Measurement:** Measuring in the computational basis yields outcome 0 with probability $|\alpha|^2$ and outcome 1 with probability $|\beta|^2$. Post-measurement state collapses to $|0\rangle$ or $|1\rangle$ accordingly.

3 The Bloch Sphere – Visualising a Qubit

Any pure qubit state can be written as

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle, \quad 0 \leq \theta \leq \pi, \quad 0 \leq \phi < 2\pi.$$

Mapping (θ, ϕ) to a point on the unit sphere gives the **Bloch sphere**.



Key points:

- $|0\rangle \leftrightarrow$ north pole ($\theta = 0$); $|1\rangle \leftrightarrow$ south pole ($\theta = \pi$).
- States on the equator ($\theta = \pi/2$) are equal superpositions (e.g. $|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$ corresponds to $\theta = \pi/2, \phi = 0$).
- Relative phase ϕ determines longitudinal position; it is irrelevant for measurement in the Z basis but crucial for interference.

4 Toy Example: Hadamard Gate Followed by Measurement

We illustrate the full cycle: state preparation $\rightarrow$ unitary evolution $\rightarrow$ measurement.

4.1 Step-by-step calculation

1. **Initial state:** $|\psi_0\rangle = |0\rangle$ ($\alpha = 1, \beta = 0$).
2. **Apply Hadamard:** $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$.

$$|\psi_1\rangle = H |0\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \equiv |+\rangle.$$

Here $\alpha = \beta = 1/\sqrt{2}$.

3. Measurement in Z basis:

$$p(0) = |\alpha|^2 = \frac{1}{2}, \quad p(1) = |\beta|^2 = \frac{1}{2}.$$

Post-measurement state is $|0\rangle$ (if outcome 0) or $|1\rangle$ (if outcome 1) with equal probability.

4.2 Numerical verification (Python-like pseudocode)

```
import numpy as np
ket0 = np.array([1,0])
H     = 1/np.sqrt(2) * np.array([[1,1],[1,-1]])
ket1 = H @ ket0           # -> [0.707..., 0.707...]
prob0 = np.abs(ket1[0])**2 # 0.5
prob1 = np.abs(ket1[1])**2 # 0.5
```

Running the snippet yields probabilities 0.5 each, confirming the analytic result.

4.3 Interpretation

- The Hadamard gate maps the state to a coherent superposition where the state has nonzero amplitudes on both computational-basis states.
- Measurement destroys the coherence, yielding a random classical bit—this is the source of quantum randomness and the foundation for algorithms like Grover’s search.

5 From Single Qubit to Multi-Qubit Systems

- The state space of n qubits is the tensor product $(\mathbb{C}^2)^{\otimes n}$, dimension 2^n .
- Entangled states (e.g. Bell state $\frac{|00\rangle + |11\rangle}{\sqrt{2}}$) cannot be written as a product of individual qubit states.
- Entanglement is a key nonclassical resource in many quantum protocols and computational models.

6 Quantum Complexity Classes

To formalise the computational advantage of quantum resources, we classify problems based on the computational resources needed to solve or verify them. When quantum resources (such as qubits, quantum gates, and measurements) are accounted for, we define new complexity classes and study their relationships with classical ones.

6.1 Key Complexity Classes

P (Polynomial Time): The class of decision problems solvable by a deterministic classical computer in polynomial time.

BPP (Bounded-error Probabilistic Polynomial-Time): The class of decision problems solvable by a probabilistic classical computer in polynomial time with an error probability of at most $1/3$ for all inputs.

BQP (Bounded-error Quantum Polynomial-Time): The class of decision problems solvable by a quantum computer in polynomial time with an error probability of at most $1/3$ for all inputs. This is the standard class representing problems that can be solved efficiently on a quantum computer.

NP (Nondeterministic Polynomial-Time): The class of decision problems for which a “yes” answer can be verified by a deterministic classical computer in polynomial time, given a polynomial-size classical proof (witness).

QMA (Quantum Merlin-Arthur): The quantum analogue of NP. It is the class of decision problems for which a “yes” answer can be verified by a quantum computer in polynomial time, given a polynomial-size quantum state as a proof (witness), with bounded error.

PSPACE (Polynomial Space): The class of decision problems solvable by a classical computer (or equivalently, a quantum computer) using a polynomial amount of memory space, with no limit on the computation time. By Savitch’s Theorem, classical deterministic and non-deterministic polynomial space are equivalent ($\text{PSPACE} = \text{NPSPACE}$).

6.2 Examples of Problems in Each Class

• P (Deterministic Polynomial Time):

- *Primality Testing:* Deciding whether a given integer is prime (solvable in polynomial time using the AKS primality test).
- *Shortest Path:* Finding the shortest path between two nodes in a graph (solvable using Dijkstra’s algorithm).

• BPP (Probabilistic Polynomial Time):

- *Polynomial Identity Testing:* Deciding whether a given multivariate polynomial (represented by an arithmetic circuit) is identically zero (efficiently solvable using the Schwartz-Zippel randomized algorithm).
- *Approximate Volume of Convex Bodies (Promise/Decision version):* Deciding whether the volume of a high-dimensional convex body is $\geq V_1$ or $\leq V_2$ (given a promise on their separation). This randomized approximation scheme runs in polynomial time, whereas deterministic algorithms require exponential time in the oracle model.

• BQP (Quantum Polynomial Time):

- *Integer Factorisation (Decision version):* Deciding whether a given integer N has a prime factor less than k (solvable in polynomial time using Shor’s algorithm, whereas the best known classical algorithm, the General Number Field Sieve, takes super-polynomial time).
- *Quantum Hamiltonian Simulation (Promise/Decision version):* Deciding whether a particular transition amplitude $\langle \phi | e^{-iHt} | \psi \rangle$ of a local Hamiltonian H is greater than a threshold A or less than B (with a promise on their separation). While this promise problem is in BQP (and can be PromiseBQP-complete), classical computers require exponential time because the state space grows as 2^n .

- **NP (Classical Verification):**

- *Boolean Satisfiability (SAT)*: Determining if there exists an assignment of truth values to variables that satisfies a given Boolean formula.
- *Travelling Salesperson Decision Problem*: Deciding if there exists a route of total cost at most K that visits all cities exactly once.

- **QMA (Quantum Verification):**

- *Local Hamiltonian Problem*: Deciding whether the ground-state (lowest eigenvalue) energy of a k -local Hamiltonian is below a threshold A or above B (with a promise that $B - A \geq 1/\text{poly}(n)$). This is the quantum analogue of the 3-SAT problem (Cook-Levin theorem).
- *Consistency of Local Density Matrices*: Deciding if a given set of low-dimensional density matrices is consistent with a single global quantum state (mixed state).

- **PSPACE (Polynomial Space):**

- *Quantified Boolean Formula (QBF)*: Deciding the truth of a fully quantified Boolean formula of the form $\forall x_1 \exists x_2 \forall x_3 \dots F(x_1, x_2, x_3, \dots)$ where F is a Boolean formula. This is the canonical PSPACE-complete problem.
- *Generalized Geography*: Deciding whether the first player has a winning strategy in a game played on a directed graph (a canonical PSPACE-complete decision problem).

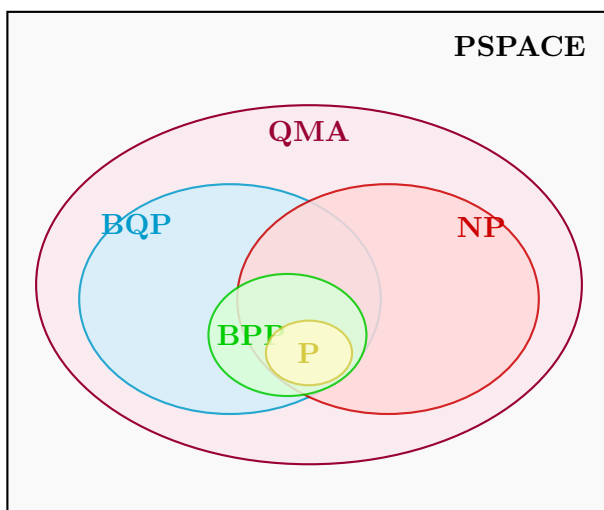
6.3 Relationships Between Complexity Classes

The standard view of relationships between these classes is illustrated by the following containment chain:

$$P \subseteq BPP \subseteq BQP \subseteq PSPACE$$

Additionally, we have:

$$NP \subseteq QMA \subseteq PSPACE \quad \text{and} \quad BQP \subseteq QMA$$



Key Insights:

1. **Quantum vs. Classical Randomness:** Because quantum computers can simulate classical probabilistic choices, we have $BPP \subseteq BQP$.
2. **BQP vs. NP:** While BQP contains problems believed to be outside P and BPP (like Integer Factorisation), it is widely believed that $NP \not\subseteq BQP$. That is, quantum computers are not expected to solve NP-complete problems efficiently.
3. **QMA as Quantum NP:** QMA generalizes NP by allowing both the proof to be a quantum state (allowing superposition) and the verifier to run a quantum algorithm.

7 Three Paradigms of Quantum Computing

While the circuit model is the most familiar representation of quantum computing, there are three primary physical and mathematical paradigms for building a universal quantum computer.

7.1 1. The Circuit Model

- **Mechanism:** Computation is represented as a sequence of discrete unitary quantum gates (such as Hadamard, CNOT, phase gates) acting on a register of initialized qubits, followed by measurement in the computational basis.
- **Dynamic:** Discrete, unitary time-evolution of state vectors.
- **Implementation:** Most superconducting qubits (IBM, Google) and trapped-ion systems (IonQ) natively adopt this gate-based model.

7.2 2. Measurement-Based Quantum Computing (MBQC)

- **Mechanism:** Also known as *one-way quantum computing*. Instead of applying gates during the computation, MBQC begins with a highly entangled multi-qubit state called a *cluster state* or *graph state* (prepared beforehand using entangling gates). The computation is carried out solely by performing sequential single-qubit measurements.
- **Dynamic:** The choice of measurement basis for a qubit depends on the outcomes of prior measurements (adaptive measurement and active feedforward). The entanglement in the cluster state is consumed as a resource, meaning the state cannot be reused (hence “one-way”).
- **Implementation:** Extremely popular in linear optical quantum computing (photonic architectures, such as PsiQuantum), where preparing large entangled states is easier than applying active entangling gates to flying photons.

7.3 3. Adiabatic Quantum Computing (AQC)

- **Mechanism:** Computation is formulated as a continuous physical process governed by the *Adiabatic Theorem*. We initialize a system in the easily-prepared ground state (lowest energy state) of an initial Hamiltonian H_{init} . We then slowly deform the system’s Hamiltonian over time $t \in [0, T]$ according to:

$$H(t) = \left(1 - \frac{t}{T}\right) H_{\text{init}} + \frac{t}{T} H_{\text{problem}}$$

If the transition is slow enough (the adiabatic condition), the system remains in its ground state. The ground state of the final Hamiltonian H_{problem} encodes the solution to the computational problem.

- **Dynamic:** Continuous-time analog evolution. The runtime T scales inversely with the square of the minimum energy gap Δ_{min} between the ground state and the first excited state.
- **Implementation:** Current quantum annealers (such as D-Wave) are generally not regarded as universal gate-model quantum computers, and universality in adiabatic settings requires stronger control assumptions (such as the ability to implement non-stoquastic entangling interaction terms).

7.4 Universality and Equivalence

Remarkably, all three paradigms are computationally equivalent. They can simulate one another with at most polynomial overhead:

1. **Circuit** $\leftrightarrow$ **MBQC**: Any circuit containing G gates on n qubits can be mapped onto a 2D cluster state of size $O(G)$ with $O(G)$ single-qubit measurements. Conversely, MBQC can be simulated in the circuit model by representing the cluster state preparation as a sequence of CNOT gates followed by measurement gates.
2. **Circuit** $\leftrightarrow$ **AQC**: Universal AQC can simulate the circuit model through the *Feynman-Kitaev history-state Hamiltonian* construction. The ground state of a history Hamiltonian is a superposition of the states at each clock cycle of the circuit:

$$|\psi_{\text{history}}\rangle = \frac{1}{\sqrt{L+1}} \sum_{t=0}^L |t\rangle_{\text{clock}} \otimes U_t \cdots U_1 |\psi_0\rangle$$

By finding the ground state of this Hamiltonian, we obtain the output of the circuit. Conversely, a quantum circuit can simulate the continuous time-evolution of AQC using Trotter-Suzuki product formulas.

3. **MBQC** $\leftrightarrow$ **AQC**: Since both can efficiently simulate and be simulated by the circuit model, they are also computationally equivalent to each other.

Thus, any algorithm (like Shor’s or Grover’s) can be run on any of the three models, and any proof of quantum speed-up in one model holds for all three.

8 Physical Realisation of Qubits

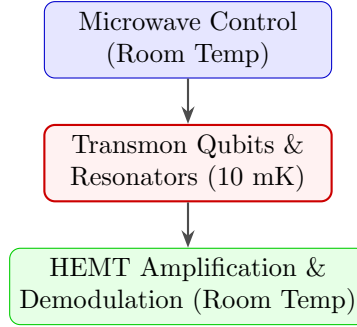
To construct a practical quantum computer, we need physical systems that can act as qubits. According to the **DiVincenzo criteria**, any physical implementation must satisfy the following requirements:

1. A scalable physical system with well-characterised qubits.
2. The ability to initialise the state of the qubits to a simple fiducial state (e.g., $|00 \dots 0\rangle$).
3. Long relevant decoherence times, much longer than the gate operation time.

4. A “universal” set of quantum gates (e.g., single-qubit rotations and a two-qubit entangling gate).
5. A qubit-specific measurement capability.

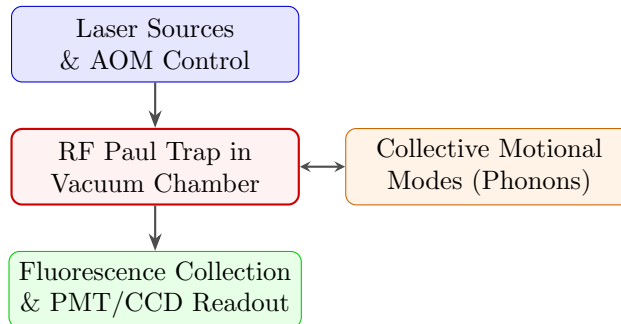
Various physical platforms are currently being developed, each exploiting different quantum systems to realise two-level states.

8.1 1. Superconducting Qubits



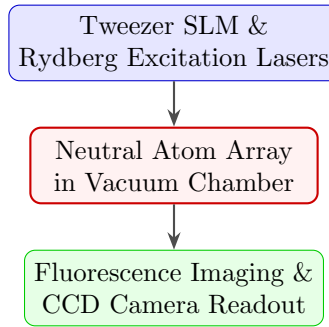
- **Physical System:** Superconducting electronic circuits fabricated on-chip, utilizing Josephson junctions (non-linear, lossless inductors) to act as artificial atoms. The most common design is the *transmon* qubit, which is engineered to reduce sensitivity to charge noise.
- **Qubit States:** The lowest two energy levels of the anharmonic oscillator circuit.
- **Control & Readout:** Single-qubit and two-qubit gates are driven by microwave pulses. Readout is achieved dispersively by measuring the transmission shift of a coupled microwave resonator.
- **Key Advantages:** Fast gate operations (nanoseconds), solid-state nature compatible with standard silicon lithography, and high scalability on-chip.
- **Primary Challenges:** Short coherence times (typically $\sim 100 \mu\text{s}$), requires sub-Kelvin cooling in dilution refrigerators ($\sim 10 \text{ mK}$), and high susceptibility to crosstalk and fabrication variations.
- **Leading Developers:** IBM, Google, Rigetti, IQM.

8.2 2. Trapped Ion Qubits



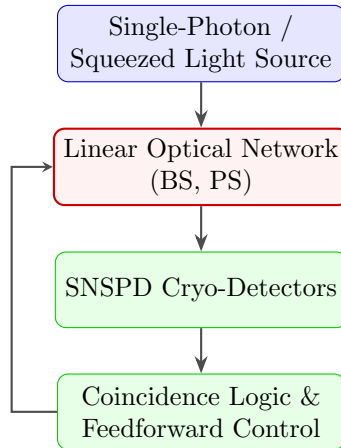
- **Physical System:** Individual charged atoms (e.g., $^{171}\text{Yb}^+$ or $^{40}\text{Ca}^+$) suspended in an ultra-high vacuum using radiofrequency electromagnetic fields (Paul traps).
- **Qubit States:** Long-lived internal electronic states, such as hyperfine ground states or stable optical metastable states.
- **Control & Readout:** Single-qubit gates are driven by lasers or microwave fields. Two-qubit gates (such as the Mølmer-Sørensen gate) exploit the collective vibrational modes (phonons) of the ion chain, coupled via Coulomb repulsion. Readout is performed via state-dependent laser fluorescence.
- **Key Advantages:** Identical, naturally perfect qubits (no fabrication variations), exceptionally long coherence times (seconds to hours), and high-fidelity gates.
- **Primary Challenges:** Slower gate speeds (microseconds to milliseconds) compared to superconducting qubits, and complex optical alignment and laser control systems. Scaling is limited by motional mode crowding in a single trap (requiring quantum charge-coupled device (QCCD) architectures or photonic interconnects).
- **Leading Developers:** Quantinuum, IonQ, Alpine Quantum Technologies.

8.3 3. Neutral Atom Qubits



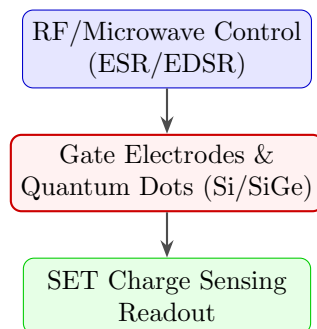
- **Physical System:** Neutral atoms (e.g., ^{87}Rb or ^{171}Yb) trapped in arrays of optical tweezers (tightly focused laser beams) in a vacuum chamber.
- **Qubit States:** Hyperfine ground states or nuclear spin states.
- **Control & Readout:** Single-qubit gates are driven by lasers or microwaves. Two-qubit gates are mediated by exciting atoms to highly excited *Rydberg states* (states with very large principal quantum number n). The strong, long-range dipole-dipole interaction between Rydberg atoms prevents simultaneous excitation of neighboring atoms (Rydberg blockade), enabling entangling gate operations. Readout is performed via fluorescence imaging on a CCD camera.
- **Key Advantages:** Highly scalable to thousands of qubits in 2D or 3D arrays, dynamically reconfigurable geometries, identical atomic qubits, and long coherence times.
- **Primary Challenges:** Relatively slow gate speeds, susceptibility to atom loss from the traps, and demanding laser power and stability requirements.
- **Leading Developers:** QuEra, Pasqal, Atom Computing.

8.4 4. Photonic Qubits



- **Physical System:** Individual quanta of light (photons) propagating through waveguides, optical fibers, or free space.
- **Qubit States:** Encoded in degrees of freedom of a single photon, such as polarization (horizontal/vertical), path (which waveguide the photon occupying), or time-bin (arrival time relative to a clock).
- **Control & Readout:** Single-qubit gates are implemented using passive linear optical components (beam splitters and phase shifters). Two-qubit gates are probabilistic because photons do not naturally interact with one another; they require auxiliary photons, measurement, and feedforward (e.g., the KLM scheme) or squeezed states in continuous-variable architectures. Readout is performed using superconducting nanowire single-photon detectors (SNSPDs).
- **Key Advantages:** Negligible decoherence at room temperature, and naturally suited for quantum communication and distributed quantum networking.
- **Primary Challenges:** Lack of direct photon-photon interactions makes two-qubit gates probabilistic, requiring massive spatial/temporal multiplexing and high-efficiency sources and detectors to overcome scaling losses.
- **Leading Developers:** PsiQuantum, Xanadu, Quandela.

8.5 5. Semiconductor Spin Qubits (Quantum Dots)



- **Physical System:** Individual electron or hole spins trapped in electrostatic potentials (quantum dots) created by metal gates on silicon or silicon-germanium (Si/SiGe) heterostructures.

- **Qubit States:** Spin-up and spin-down states of the trapped particle in an external magnetic field.
- **Control & Readout:** Single-qubit gates are driven by electron spin resonance (ESR) or electric dipole spin resonance (EDSR). Two-qubit gates are implemented via the short-range Heisenberg exchange interaction between adjacent quantum dots. Readout is achieved using gate-based reflectometry or charge sensors.
- **Key Advantages:** Extremely small physical size (sub-micron footprint), and potential to leverage the existing multi-billion-dollar CMOS semiconductor manufacturing industry.
- **Primary Challenges:** High sensitivity to local charge noise and magnetic noise from the host lattice, extremely high fabrication precision requirements, and operation at sub-Kelvin temperatures (~ 100 mK).
- **Leading Developers:** Intel, Diraq, Quantum Motion.

9 Discussion Problems / Exercises

1. **Interference and Qubit Phases:** Suppose a single qubit is prepared in the state:

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle.$$

- (a) Show that if we measure $|\psi\rangle$ in the computational (Z) basis, the outcome probabilities $P(0)$ and $P(1)$ are completely independent of the relative phase ϕ .
- (b) Suppose we first apply a Hadamard gate H to $|\psi\rangle$ and then perform a Z -basis measurement. Calculate the probability of obtaining the outcome 0, and show that it explicitly depends on ϕ . Why is this relative phase dependency critical for quantum computation?

Takeaway for Session: This exercise demonstrates quantum interference. It shows that while phase information is invisible in a direct measurement, unitary transformations (like the Hadamard gate) can map phase differences to measurable amplitude differences. Detailed proof is provided in Appendix A.

2. **Why Grover's Search Does Not Put NP in BQP:** Grover's unstructured search algorithm finds a target state in a space of size $N = 2^n$ using $O(\sqrt{2^n})$ queries.
 - (a) Explain the difference between polynomial-time complexity (required for BQP) and Grover's query complexity.
 - (b) Why does the existence of Grover's search algorithm not imply that $\text{NP} \subseteq \text{BQP}$, despite NP containing many search problems?

Takeaway for Session: Grover's algorithm highlights that quantum computers are not magic search engines that can solve NP-complete problems instantly. An exponential search space still requires exponential time, though with a quadratic query speedup. Detailed explanation is provided in Appendix A.

3. **Simple Two-Qubit Cluster State (MBQC):** In MBQC, a two-qubit cluster state is prepared by initializing two qubits in the $|+\rangle$ state and applying a Controlled-Z (CZ) gate:

$$|\psi_{12}\rangle = \text{CZ}(|+\rangle \otimes |+\rangle).$$

- (a) Write down the state $|\psi_{12}\rangle$ in the standard computational basis ($|00\rangle, |01\rangle, |10\rangle, |11\rangle$).
- (b) Show that $|\psi_{12}\rangle$ is an entangled state (i.e., it cannot be written as a product state $|\phi_1\rangle \otimes |\phi_2\rangle$).

Takeaway for Session: This exercise provides a concrete example of preparing a simple entangled state. Entanglement cannot be factored into independent single-qubit states, illustrating the non-local correlation that underpins multi-qubit systems. Detailed derivation is provided in Appendix A.

10 Take-away Messages

1. A qubit lives in a two-dimensional complex vector space; its state is described by two complex amplitudes.
2. Unitary gates rotate the Bloch-sphere vector; measurement projects onto the computational basis.
3. Simple gates (Hadamard, Pauli) already illustrate non-classical behaviour: superposition and interference.
4. Scaling to many qubits leads to an exponentially large Hilbert space, the bedrock of quantum advantage.

A Detailed Solutions to Discussion Problems

1. Interference and Qubit Phases Solution:

(a) Measuring in the Z -basis projects onto the basis states $\{|0\rangle, |1\rangle\}$. The probabilities are:

$$P(0) = |\langle 0 | \psi \rangle|^2 = \left| \cos \frac{\theta}{2} \right|^2 = \cos^2 \frac{\theta}{2}$$

$$P(1) = |\langle 1 | \psi \rangle|^2 = \left| e^{i\phi} \sin \frac{\theta}{2} \right|^2 = |e^{i\phi}|^2 \sin^2 \frac{\theta}{2} = \sin^2 \frac{\theta}{2}$$

Since $|e^{i\phi}|^2 = 1$ for any real ϕ , the probabilities do not depend on ϕ .

(b) Applying the Hadamard gate:

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Thus, the state after the Hadamard gate is:

$$H|\psi\rangle = \cos \frac{\theta}{2} \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) + e^{i\phi} \sin \frac{\theta}{2} \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

Grouping the terms for $|0\rangle$ and $|1\rangle$:

$$H|\psi\rangle = \frac{1}{\sqrt{2}} \left(\cos \frac{\theta}{2} + e^{i\phi} \sin \frac{\theta}{2} \right) |0\rangle + \frac{1}{\sqrt{2}} \left(\cos \frac{\theta}{2} - e^{i\phi} \sin \frac{\theta}{2} \right) |1\rangle$$

The probability of measuring 0 is:

$$P(0) = |\langle 0 | H|\psi\rangle|^2 = \frac{1}{2} \left| \cos \frac{\theta}{2} + e^{i\phi} \sin \frac{\theta}{2} \right|^2$$

Expanding using $|z|^2 = zz^*$:

$$\begin{aligned} P(0) &= \frac{1}{2} \left(\cos \frac{\theta}{2} + e^{i\phi} \sin \frac{\theta}{2} \right) \left(\cos \frac{\theta}{2} + e^{-i\phi} \sin \frac{\theta}{2} \right) \\ &= \frac{1}{2} \left(\cos^2 \frac{\theta}{2} + \sin^2 \frac{\theta}{2} + (e^{i\phi} + e^{-i\phi}) \cos \frac{\theta}{2} \sin \frac{\theta}{2} \right) \\ &= \frac{1}{2} \left(1 + 2 \cos \phi \cos \frac{\theta}{2} \sin \frac{\theta}{2} \right) \\ &= \frac{1}{2} (1 + \sin \theta \cos \phi) \end{aligned}$$

where we used the identity $\sin \theta = 2 \sin(\theta/2) \cos(\theta/2)$. This probability explicitly depends on the relative phase ϕ .

2. Why Grover's Search Does Not Put NP in BQP Solution:

(a) For a problem to be in BQP, it must be solvable in polynomial time, meaning the number of gates (or queries) scales as $O(n^c)$ for some constant c , where n is the input size (number of qubits). Grover's algorithm requires $O(\sqrt{2^n}) = O(2^{n/2})$ queries. Because $2^{n/2}$ scales exponentially with n , Grover's query complexity is still super-polynomial (exponential) in the size of the input.

- (b) The class NP contains decision versions of search problems (e.g., finding a satisfying assignment to a formula or a path of a certain length). While Grover's search can search the state-space of potential witnesses, doing so requires $O(2^{n/2})$ steps. Since this running time is exponential, it does not provide a polynomial-time solution for NP-complete problems. Thus, Grover's algorithm does not show that $\text{NP} \subseteq \text{BQP}$. In fact, it is known that in a black-box oracle model, Grover's $O(\sqrt{N})$ query complexity is the absolute best possible quantum bound (Bennett, Bernstein, Brassard, and Vazirani, 1997).

3. Simple Two-Qubit Cluster State (MBQC) Solution:

- (a) First, we expand the initial state:

$$|+\rangle \otimes |+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}} = \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle).$$

The Controlled-Z gate acts on the computational basis as $\text{CZ}|x, y\rangle = (-1)^{xy}|x, y\rangle$. Thus, it applies a phase of -1 only to the state $|11\rangle$:

$$|\psi_{12}\rangle = \text{CZ}(|+\rangle |+\rangle) = \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle - |11\rangle).$$

- (b) Suppose $|\psi_{12}\rangle$ could be factored as a product state of two single-qubit states:

$$|\phi_1\rangle = a|0\rangle + b|1\rangle, \quad |\phi_2\rangle = c|0\rangle + d|1\rangle.$$

Their product state is:

$$|\phi_1\rangle \otimes |\phi_2\rangle = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle.$$

For this to equal $|\psi_{12}\rangle$, the coefficients must satisfy:

$$ac = \frac{1}{2}, \quad ad = \frac{1}{2}, \quad bc = \frac{1}{2}, \quad bd = -\frac{1}{2}.$$

From the first two equations, if $ac = 1/2$ and $ad = 1/2$, then $c = d$ (since $a \neq 0$ because $ac \neq 0$). However, if $c = d$, then $bc = bd$. But the third and fourth equations state that $bc = 1/2$ and $bd = -1/2$, which is a contradiction ($1/2 = -1/2$). Thus, no such single-qubit states exist, proving that $|\psi_{12}\rangle$ is entangled.

Further reading (optional)

- **Textbooks and Lecture Notes:**

- Nielsen & Chuang, *Quantum Computation and Quantum Information*, Chapter 2 (foundations), Chapter 3 (complexity theory), and Chapter 4 (circuits).
- John Preskill, *Lecture Notes for Physics 219: Quantum Computation*, Chapters 1 & 2.
- Scott Aaronson, *Quantum Computing Since Democritus*, Cambridge University Press, Chapters 10 (Quantum Complexity) and 14 (BQP and NP).

- **Alternative Paradigms Literature:**

- Raussendorf, R., & Briegel, H. J. (2001). “A One-Way Quantum Computer.” *Physical Review Letters*, 86(22), 5188 (Foundational paper for MBQC).
- Farhi, E., et al. (2000). “Quantum Computation by Adiabatic Evolution.” *arXiv:quant-ph/0001106* (Foundational paper for AQC).
- Aharonov, D., et al. (2007). “Adiabatic Quantum Computation is Equivalent to Standard Quantum Computation.” *SIAM Journal on Computing*, 37(1), 166-194 (Proof of AQC universality).

Acronyms

Acronym	Full Description
AOM	Acousto-Optic Modulator
AQC	Adiabatic Quantum Computing
BPP	Bounded-error Probabilistic Polynomial-time
BQP	Bounded-error Quantum Polynomial-time
BS	Beamsplitter
CCD	Charge-Coupled Device
CMOS	Complementary Metal-Oxide-Semiconductor
CZ	Controlled-Z
EDSR	Electric Dipole Spin Resonance
ESR	Electron Spin Resonance
HEMT	High-Electron-Mobility Transistor
KLM	Knill-Laflamme-Milburn (scheme for linear optics QC)
MBQC	Measurement-Based Quantum Computing
NP	Nondeterministic Polynomial-time
P	Polynomial-time
PMT	Photomultiplier Tube
PS	Phase Shifter
PSPACE	Polynomial Space
QBF	Quantified Boolean Formula
QCCD	Quantum Charge-Coupled Device
QMA	Quantum Merlin-Arthur
RF	Radio Frequency
SAT	Boolean Satisfiability Problem
SET	Single-Electron Transistor
SLM	Spatial Light Modulator
SNSPD	Superconducting Nanowire Single-Photon Detector
TSP	Travelling Salesperson Problem